

SAHIL

Software Engineer | Final Year Undergrad, DJSCE Mumbai

Profile Summary: Adaptable and AI-forward Software Engineer with a proven track record of architecting and deploying complex systems. Demonstrated expertise across diverse tech stacks (Python, Java, C#, Flutter) and domains, including Hybrid RAG systems, Infrastructure-as-Code orchestration, high-throughput ETL pipelines, and scalable cloud deployments (GCP/GKE). Proficient in integrating LLMs/SLMs to build intelligent, context-aware applications.

Professional Experience (JM Financial Internship)

Enterprise Financial Regulatory RAG Platform

Full-Stack Development | Architecture & UI Implementation

Tech: Python, FastAPI, Flutter, SQLite, ChromaDB, Gemini, EventBus, Docker

Architected and developed a comprehensive Hybrid Retrieval-Augmented Generation (RAG) system to process and query complex financial regulatory documents (SEBI, NSE, BSE).

- **3-Tier RAG Engine:** Built a vector-less primary semantic path utilizing a PageIndex JSON tree for document-structure-aware navigation, backed by an LLM agent that autonomously traverses document sections and auto-follows internal references. Engineered a tiered fallback system (Tier 1: Context Cache, Tier 2: Vector Search via ChromaDB, Tier 3: PageIndex Agent).
- **Zero-LLM Structured Path:** Implemented an automated table extraction pipeline mapping regulatory data to an SQLite RowStore, enabling deterministic, sub-millisecond queries (COUNT, LIST, EXPORT) without LLM latency or hallucinations.
- **Advanced Flutter UI:** Built the entire client-side interface managing distinct operational modes (RAG Chat, Due Diligence/VDR runs, and Template Generation). Handled complex UI state management, background SSE (Server-Sent Events) streaming, session recovery loops, and workspace isolation using Riverpod and robust closure patterns.
- **Intent Classification & Routing:** Designed a multi-stage intent router combining deterministic regex rules and SLM (Small Language Model) arbitration to strictly gate structured queries versus semantic exploration.

Compliance Audit Workflow Engine

Backend Engineering | Enterprise Workflows

Tech: Java, Spring Boot, Custom '#' Schema

Developed a live, visual workflow engine utilized by the Compliance team to manage complex audit pipelines.

- Engineered an "n8n-like" dynamic workflow system featuring custom nodes and a manually created '#' based schema for extreme flexibility.
- Implemented secure attachment handling and audit trails, allowing items and news to flow hierarchically from junior analysts to senior management.
- Integrated granular permission controls, enabling assignees to push items forward or trigger targeted revision loops.

High-Volume ETL Pipeline Migration

Data Engineering | C# to PostgreSQL

Tech: C#, PostgreSQL, Multi-threading

Led the conversion of legacy C# data scripts to optimized PostgreSQL environments to handle massive financial datasets.

- Processed millions of rows of financial data with strict consistency guarantees.
- Implemented advanced database optimization techniques including multi-threading, data sharding, and connection pooling to ensure high-throughput execution at scale.
- Resolved numerous edge cases related to historical data anomalies and legacy format inconsistencies.

JM PRO App - Dynamic Notification Engine

Backend Engineering | Automation

Tech: Python, Web Scraping, Financial APIs, GCP VMs, Cron

Built a real-time, dynamic notification engine driving the JM PRO mobile application.

- Engineered scraper bots and integrated external Finance APIs to continuously update company databases with real-time market data.
- Scheduled and dispatched highly personalized push notifications including IPO alerts, portfolio summaries, research report drops, and client birthday greetings.
- Deployed the engine on Google Cloud Platform (GCP) Virtual Machines, orchestrated via Cron jobs for reliable, time-sensitive delivery.

Cloud Infrastructure & CI/CD Operations

DevOps | Cloud Engineering

Tech: Google Cloud Platform (GCP), GKE, Cloud Run, Artifact Registry, Docker

Actively contributed to the team's DevOps lifecycle, containerizing applications and managing cloud deployments.

- Containerized backend services and scripts using Docker, pushing images to GCP Artifact Registry as part of automated CI/CD pipelines.
- Managed deployments across Google Kubernetes Engine (GKE) and Cloud Run, ensuring high availability and scalability for backend services.
- Configured Cloud Storage (Buckets) and managed automated VM scheduling scripts.

Independent Projects

SimArch - Visual IaC Orchestrator

Personal Project | Infrastructure Simulator

Tech: Flutter, Python, FastAPI, Docker SDK, MongoDB, EventBus

Created a comprehensive, drag-and-drop local Infrastructure-as-Code (IaC) orchestrator and architecture simulator, allowing users to design distributed systems on a canvas and bring them to life instantly.

- **Dual Execution Modes:** Built a toggleable engine supporting both "Simulate" (Python state machines mimicking container lifecycles) and "Execute" (provisioning real Docker containers on the host machine).
- **Real Infrastructure Provisioning:** Automated the deployment and networking of Postgres, Redis, MinIO (S3), RabbitMQ, Nginx Load Balancers, and Prometheus/Grafana monitoring stacks directly from the UI.
- **Live Telemetry:** Engineered a WebSocket-based `tty_burst` system to stream live ANSI-colored terminal output from all active nodes directly into the Flutter frontend.
- **Advanced Architecture:** Utilized a FastAPI backend with MongoDB persistence, an internal DAG execution gate to respect upstream dependencies, and dynamic Docker bridge networks for cross-container communication.

Technical Arsenal & Capabilities

Languages & Frameworks • Python (FastAPI, Pandas, Scripting) • Dart (Flutter 3.x, Riverpod) • Java (Spring Boot) • C# (.NET)	Cloud & Infrastructure • Docker, Kubernetes (GKE) • Google Cloud (Cloud Run, VMs, Storage, Artifact Registry) • PostgreSQL, MongoDB, Redis, RabbitMQ • Nginx, Prometheus, Grafana
AI & Machine Learning • Generative AI Integration (Gemini, Anthropic) • Local SLMs / LLMs via Ollama • Agentic workflows & Prompt Engineering • Retrieval-Augmented Generation (RAG) Architecture	Core Competencies • High-Volume Data Engineering (ETL) • System Design & Architecture Simulation • Rapid cross-stack adaptability • CI/CD pipeline automation